

ARUNKUMAR K R

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🐙 GitHub | 🏆 Leetcode | 🏠 Hackerrank

SUMMARY

AI Engineer and Backend Systems Developer specializing in end-to-end LLM application development, computer vision pipelines and production-grade API systems. Experienced in building multi-modal AI products using Python, FastAPI, LangChain, and cloud-hosted models. Targeting AI and ML roles where I can architect intelligent systems and the scalable infrastructure around them. Published researcher in deep learning-based industrial inspection (YOLOv8), RHCSA-certified, and competitive programmer with 500+ DSA problems solved.

EDUCATION

KPR Institute of Engineering and Technology, Coimbatore	Sept 2023 – Apr 2027
B.E. Computer Science and Engineering	CGPA: 8.77 / 10
Kongu Vellalar Matriculation Higher Secondary School, Perundurai	Jun 2022 – Mar 2023
Higher Secondary Education (Class XII)	94.6%

SKILLS

- **AI & Machine Learning:** LangChain, TensorFlow, PyTorch, Hugging Face, ChromaDB, RAG and LLM pipelines
- **Backend Development:** FastAPI, Node.js, Express.js, Django, REST APIs
- **Database & Infrastructure:** PostgreSQL, MongoDB, Redis, Docker, Linux, Git
- **Languages:** Python, Java, JavaScript, C, SQL
- **Soft Skills:** Problem Solving, Team Collaboration, Technical Communication, Leadership

EXPERIENCE

Reccsar Pvt Ltd	Jan 2025 – Feb 2025
<i>Backend Developer Intern</i>	
• Built and shipped 5+ production features using React.js, Node.js, and Express.js, improving operational workflows for 100+ active users. • Integrated 10+ REST APIs with a PostgreSQL backend, reducing manual data entry effort by ~30% .	

PROJECTS

Prism AI – Multi-Modal AI Content Platform	Live Demo GitHub
<i>Python, FastAPI, Groq LLM, Hugging Face FLUX.1, async APIs</i>	
• Engineered a production-deployed multi-modal AI SaaS that generates SEO blog posts, video scripts, and images using Groq LLM and Hugging Face FLUX — supporting concurrent AI generation tasks with async architecture. • Unified all AI inference services and frontend under a single FastAPI backend with tiered pricing, JWT authentication, and multi-language support, serving real users at scale.	
ProManage – Scalable Project Management API	Live Demo GitHub
<i>Node.js, TypeScript, PostgreSQL, Redis, JWT</i>	
• Architected a REST API with Role-Based Access Control and JWT authentication, supporting distinct permission tiers for admins, managers, and members across concurrent teams. • Reduced API response time by 45% using Redis caching and Node.js Streams, sustaining low latency under 500+ concurrent requests.	
RouteX – Intelligent Document Classification Pipeline	Live Demo GitHub
<i>Python, PyMuPDF, Tesseract OCR, NLP, TF-IDF</i>	
• Built an end-to-end NLP pipeline using OCR and text classification to automatically route both scanned and digital PDFs with 95% accuracy. • Implemented auto-tagging via NLP summarization and TF-IDF keyword extraction, eliminating manual categorization for large document batches.	

RESEARCH & ACHIEVEMENTS

- **Research Publication:** "Real-Time Surface Defect Detection in Metals using YOLOv8" – presented and published at the **3rd ICEAMST International Conference**; applied deep learning for industrial computer vision.
- **RedHat Certified System Administrator (RHCSA)** – Recognized Linux system administration certification.
- **Competitive Programming:** Solved **500+ DSA problems** on LeetCode and HackerRank; certified in Problem Solving (Intermediate), Python, Java and SQL.
- **Leadership:** Executive Member, Dept. Association – organized technical events and workshops for 300+ students.